

Mukharbek Organokov | AI Engineer

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Physicist-turned-ML-engineer with 7+ years shipping production machine learning - multi-agent LLM orchestration, large-scale forecasting, and structured-data prediction at startup and enterprise scale. Moves quickly on ambiguous, open-ended problems, owning them end-to-end from research question to deployed system, and shaping both product and technical direction in early-stage teams.

Professional skills: Machine Learning | Deep Learning | Forecasting | Uncertainty Quantification
Fine-tuning | Feature engineering | Knowledge Graphs | GNN | NLP | LLM | RAG
Problem-Solving | Leadership | Project Management | Excellent Communication

Programming skills: **Proficient:** Python, SQL, Bash | **Working knowledge:** Rust | **Familiar:** Go, C/C++

Technical skills: PyTorch | Keras | XGBoost | Scikit-learn | Scikit-optimize | Ray | Optuna
NumPy | Pandas | SciPy | NetworkX | Jupyter | Asyncio | Git | GitHub Actions
Docker | Kubernetes | Temporal | AWS | GCP | gRPC | Kafka | Helm | ArgoCD
Kubeflow | MLFlow | KServe | vLLM | ONNX | FastAPI | Vector DBs | LangChain

Work experience

- 🇫🇷 **Senior Machine Learning Engineer @ Temelion** 12.2025 - ... now
- Architected a doc intelligence service that parses, structures, and extracts knowledge from construction specs.
 - Engineered a defense-in-depth LLM extraction pipeline for public procurement scoring grids with robust QA validation.
 - Built and shipped a zero-shot doc classifier to production, enabling automatic categorization of construction documents.
 - Designed and integrated a high-performance OCR system, improving document processing speed by 90%
 - Delivered an end-to-end PDF-to-Excel transformation pipeline for construction pricing documents.
 - Introduced MLflow and Temporal workflow orchestration to the team, establishing reproducible AI evaluation.
- 🇫🇷 **Senior AI Stack Engineer @ FlexAI** 09.2025 - 12.2025
- Architected and implemented Go-based KServe integration within AI core infrastructure to deliver scalable inference.
 - Prototyped a Rust-based observability platform for AI core infrastructure.
 - Integrated several MLOps tools into backend services to enable automated training.
- 🇫🇷 **Senior Machine Learning Engineer @ Cisco Systems** 03.2023 - 08.2025
- Developed a cybersecurity tool for the real-time detection of cyber threats in cloud-native applications (**patent**).
 - Engineered a two-stage ML pipeline (PyTorch + XGBoost) on tabular telemetry to boost detection accuracy (**patent**).
 - Leveraged time-series modeling and forecasting to improve real-time detection of cyber attacks (**patent**).
 - Built an IT network-management automation tool using GNNs, LLMs, and Knowledge Graphs (and Batfish).
 - Designed and implemented a multi-agent LLM system for automated network reasoning.
 - Established and led the MLOps process using Kubeflow to monitor and maintain machine learning models lifecycle.
 - Designed a robust and highly scalable inference service using KServe that offers serverless inferencing.
 - Integrated Kafka data streams with the KServe inference service into the internal APIs and deployed.
- 🇫🇷 **Machine Learning Ops Engineer @ Integral Ad Science** 07.2022 - 03.2023
- Implemented a benchmarking tool to support the team to track pipeline metrics and evaluate the quality of models.
 - Extended the tool for custom decision rules evaluation based on IAB taxonomy and GARM standards.
 - Empowered in scaling of inference system to enable real-time processing of millions of videos/texts (YouTube, Twitter).
- 🇫🇷 **Data Scientist Engineer @ sense4data** 07.2020 - 07.2022
- Built a computer vision tool for the detection and classification of defects in railway ties for Roav7.
 - Developed fraud detection tool for OneSpan and Deutsche Bank to reduce their financial losses.
 - Improved real-time detection of complex fraud using temporal and behavioral pattern analysis.
 - Designed and integrated a machine learning pipeline to forecast match outcomes in various sports.
 - Increased ROI of forecasting model to positive, producing a total return ~3%.

- 🇫🇷 Mentor in Data Science @ Kaggle BIPOC Program** Mar. 2021 - May. 2021
 ◦ Taught basic deep learning methods to a mentee from Nigeria.
- 🇷🇺 Teaching Assistant @ SPb Polytechnic University** Sep. 2020 - Dec. 2020
 ◦ Taught master students the Machine Learning Crash Course (MLCC).
- 🇬🇧 Data Science Pet-Projects @ The Alan Turing Institute** Sep. 2019 - Dec. 2019
 ◦ 2025 Collaborated on: "Detecting Shallow Gas from Marine Seismic Images.", DOI:10.5281/zenodo.15728505
 ◦ 2021 Co-facilitated: "Multi-sensor based intelligent machining process monitoring.", DOI:10.5281/zenodo.7075712
 ◦ 2019 Collaborated on: "Semantic and Instance Segmentation of 3D Point Clouds.", DOI:10.5281/zenodo.3878499
 ◦ 2019 Collaborated on: "Understanding the influences of network measures on customer perception."
- 🇫🇷 Data Science Pet-Projects @ Kaggle (Google)** Sep. 2019 - Aug. 2021
- | Rank [Best Global: 143rd] | Tech Stack | |
|---|--------------------------|--------|
| ◦ 34th [Top 4%]: M5 Forecasting - Uncertainty | Keras LightGBM | silver |
| ◦ 42th [Top 9%]: March Machine Learning Mania 2021 | Scikit-Learn LightGBM | silver |
| ◦ 76th [Top 5%]: Jigsaw Multi-Lang. Toxic Comment Classification | TensorFlow XLM-RoBERTa | silver |
| ◦ 87th [Top 2%]: Mechanisms of Action (MoA) Prediction | PyTorch LightGBM | silver |
| ◦ 88th [Top 7%]: Parkinson's Freezing of Gait Prediction | PyTorch LightGBM | bronze |
| ◦ 95th [Top 6%]: AMP®-Parkinson-s Disease Progression Prediction | PyTorch LightGBM | bronze |
| ◦ 95th [Top 6%]: Understanding Clouds from Satellite Images | PyTorch Albumentations | bronze |
| ◦ 99th [Top 4%]: Predicting Molecular Properties | Keras LightGBM | silver |
- 🇫🇷 PhD Researcher @ IPHC (CNRS)** Jan. 2016 - Aug. 2019
 ◦ Provided statistical and data analysis of astrophysical data using C/C++ and Python.
- 🇫🇷 Graduate Student @ IRFU (CEA-Saclay) & Ecole Polytechnique** Feb. 2015 - Jul. 2015
 ◦ Provided statistical and data analysis of astrophysical data using C/C++ and ROOT.
- 🇷🇺 Graduate Student and Research Assistant @ Ioffe Institut** Feb. 2012 - Jul. 2013
 ◦ Carried out Monte-Carlo simulations using C/C++, MATLAB, and Geant4.

Education

🇫🇷 Ph.D. in Astrophysics	Université de Strasbourg	2019
🇫🇷 M.Sc. in High Energy Physics	École Polytechnique (Université Paris-Saclay)	2015
🇷🇺 M.Sc. in Financial Management	St.Petersburg State University	2014
🇷🇺 M.Sc. in Space Physics	St.Petersburg State Polytechnical University	2013
🇷🇺 B.Sc. in Physics	Kabardino-Balkarian State University	2011

Selected Trainings

🇫🇷 GenAI Blue Belt, virtual, Cisco	2024
🇫🇷 OWASP Top 10 Vulnerability Risks and How to Mitigate Them, virtual, OWASP	2023
🇬🇧 OxML: Oxford Machine Learning Summer School, virtual, Oxford	2021
🇨🇭 Seventh Machine Learning in High Energy Physics Summer School, virtual, EPFL	2021
🇬🇧 Astro Hack Week: Machine Learning for Astronomy, KICC, University of Cambridge	2019
🇵🇱 ISSDL: International Summer School on Deep Learning, Gdańsk University of Technology	2019
🇩🇰 Advanced Topics in Machine Learning. Danmarks Tekniske Universitet	2018
🇨🇭 Ifl Summer School on Machine Learning, Universität Zürich	2018
🇫🇷 Advanced Software Programming for Astrophysics & Astroparticles, LAPP, Annecy-le-Vieux	2017

Miscellaneous

[in](https://www.linkedin.com/in/circassia/) www.linkedin.com/in/circassia/
[G](https://github.com/kabartay) https://github.com/kabartay
[k](https://www.kaggle.com/muhakabartay) www.kaggle.com/muhakabartay
[T](https://twitter.com/circassia_ai) https://twitter.com/circassia_ai

Residence: 🇫🇷 | Work permission: 🇫🇷
 Citizenship: 🇷🇺
 Languages: 🇬🇧 fluent | 🇫🇷 basic | 🇷🇺 fluent | 🇩🇪 fluent
 Google Scholar: jYZaDVoAAAAJ